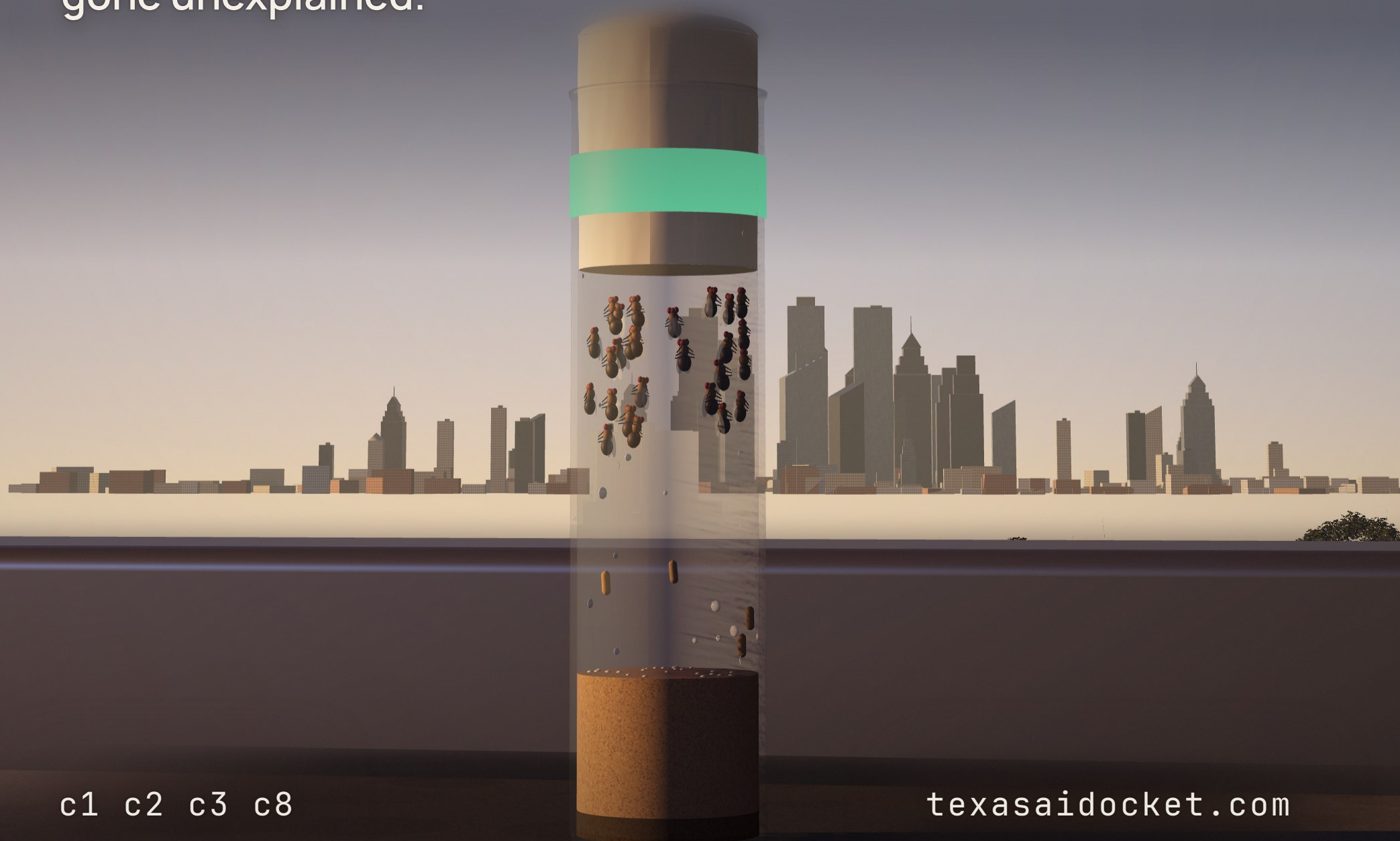


Software pointed at a gene. Flies ran the test.

Researchers at Texas Children's and Baylor name variants in BRSK1 as a likely diagnosis for a rare disorder that had gone unexplained.



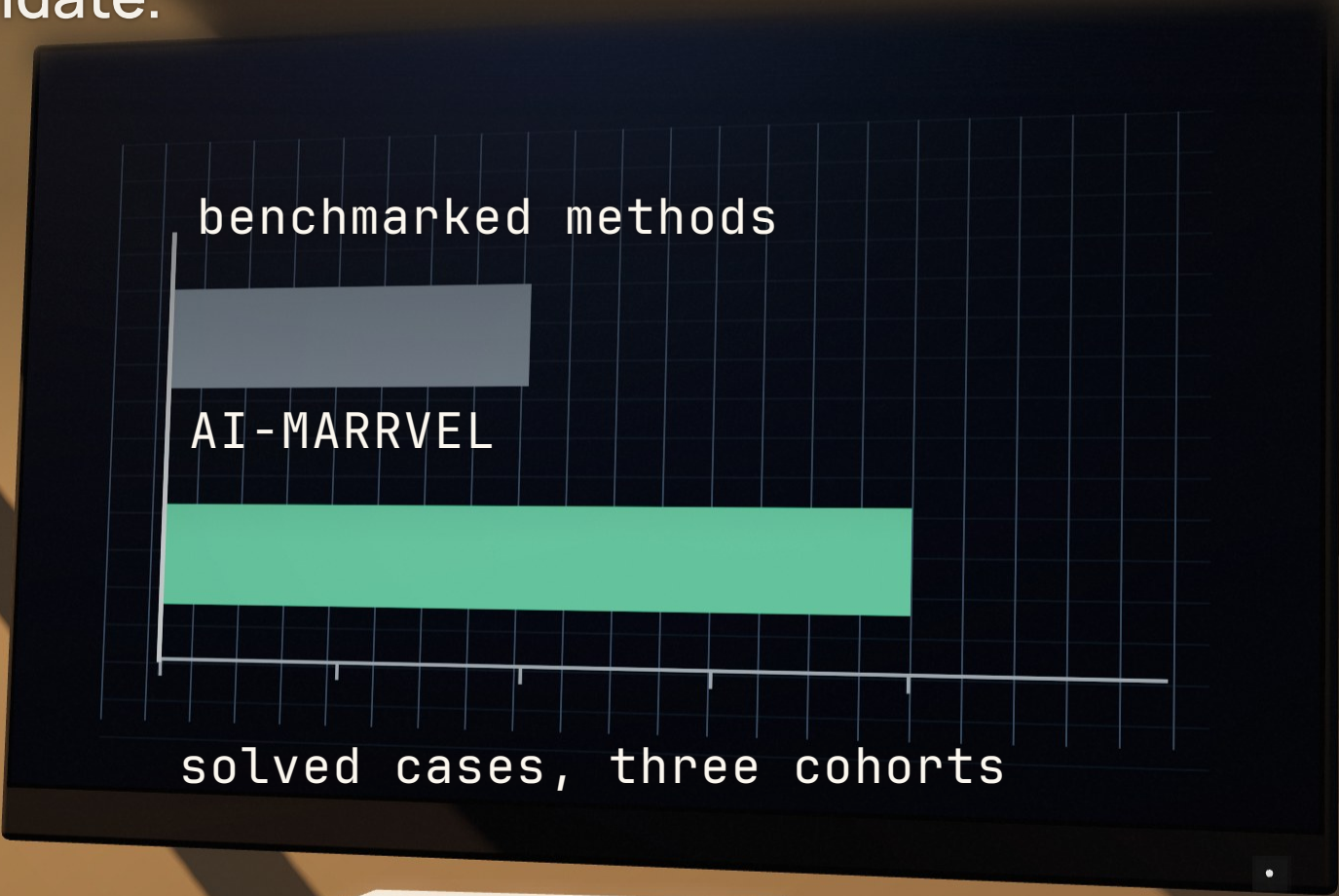
One child, and no answer.

The child was enrolled in the Texome Project, which offers free genetic testing to medically underserved people with rare, undiagnosed conditions. Standard analysis of the child and a parent did not reveal an answer.



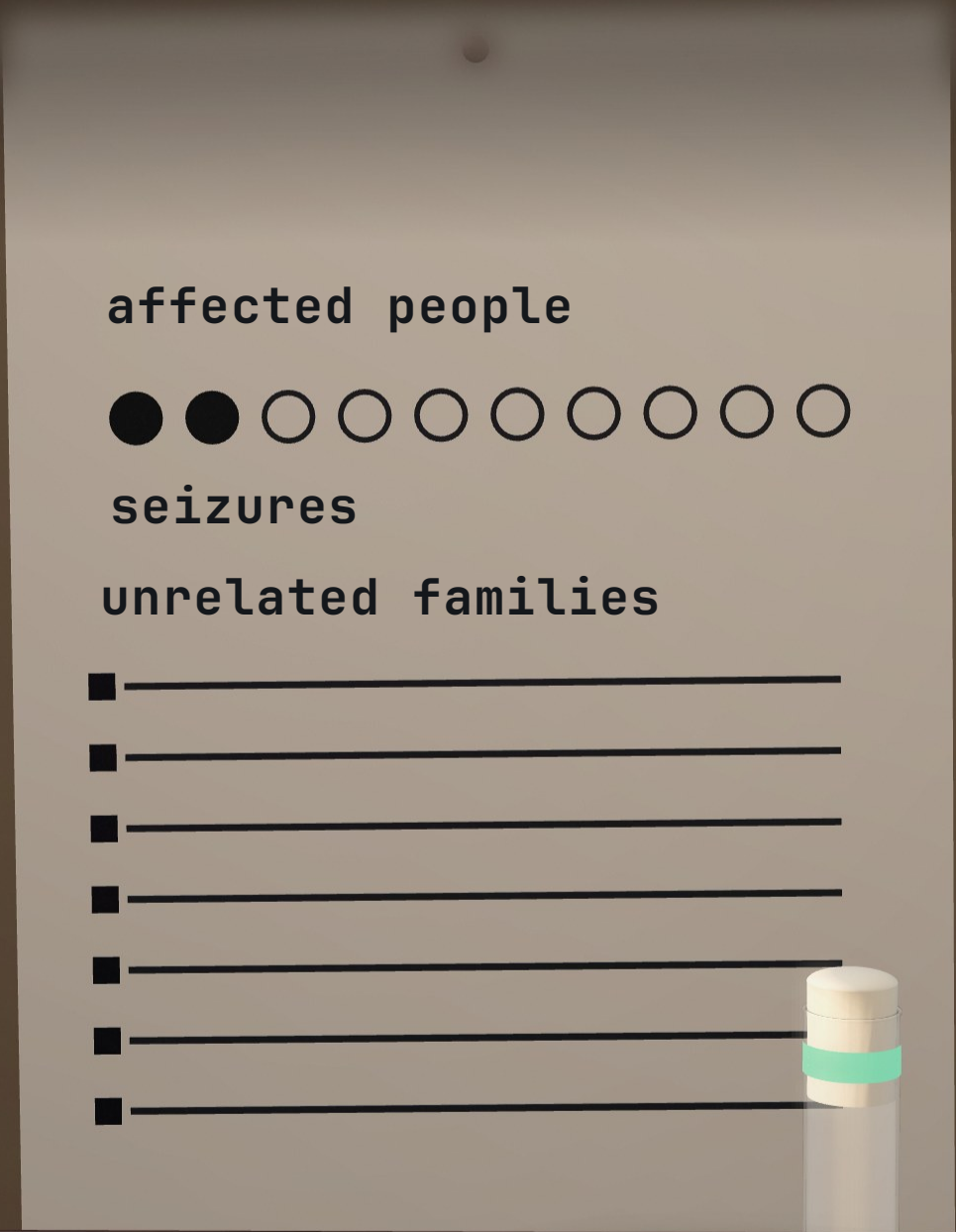
Then software flagged BRSK1.

AI-MARRVEL, a classifier trained on over 3.5 million variants, highlighted a rare change in BRSK1 as a promising candidate.



Ten people. Seven unrelated families.

Shared through GeneMatcher, the finding drew in other families with rare variants in the same gene. Two of the ten had seizures.



A good pointer is still a pointer.

On one confidence metric, AI-MARRVEL reached 98% precision and found 57% of diagnosable cases among 871. That is not its accuracy.

98% precision

57% of diagnosable cases found

With the gene off, the flies had trouble moving.

The fly version of BRSK1 is called sugar-free frosting. Disabled, it also left the flies with shorter lives.



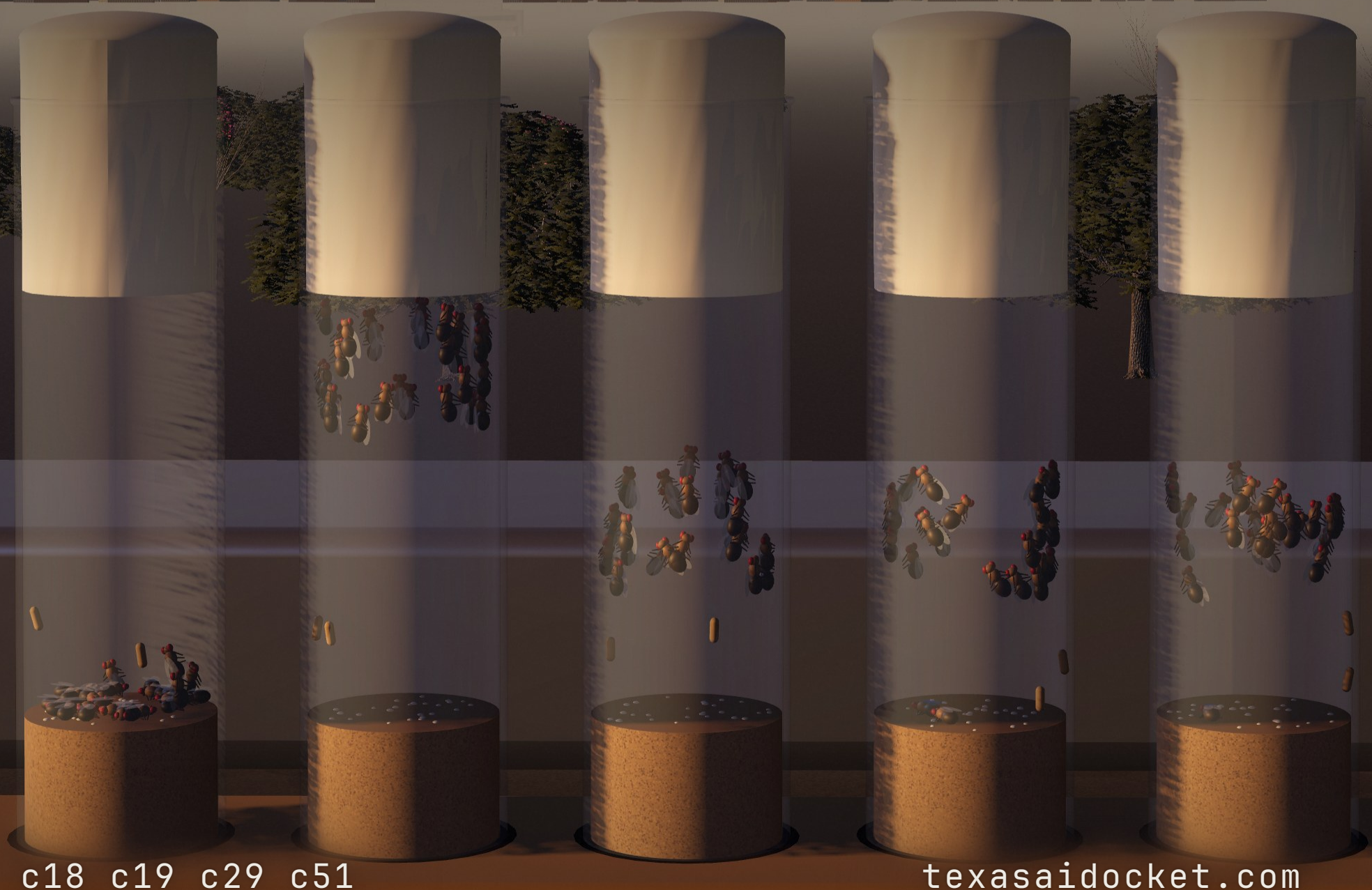
The patients' versions only partly fixed the flies.

The normal human gene largely corrected the flies. The team modeled three patient variants, and those restored the flies only partially.

gene off

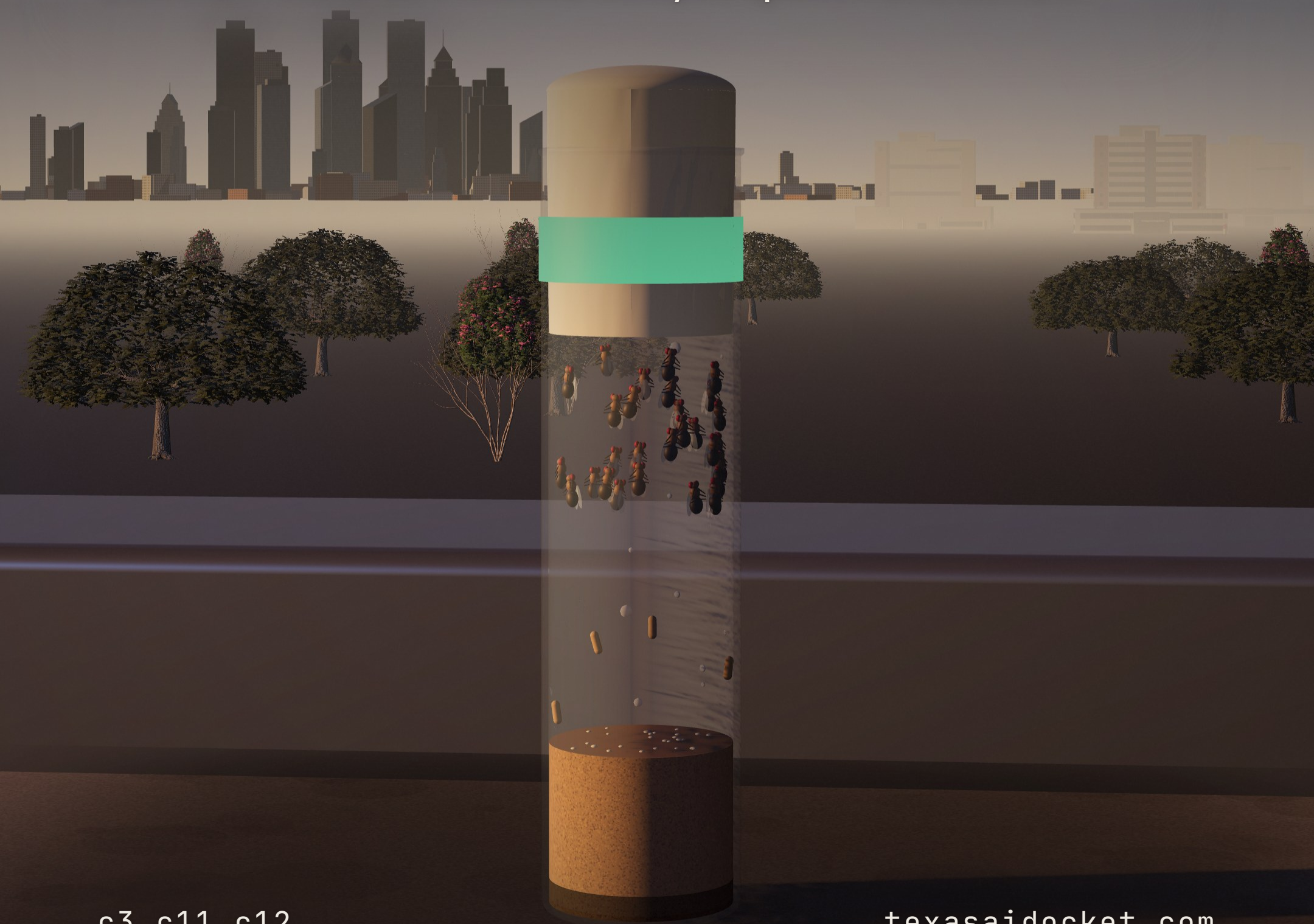
human gene

patient variants



A likely diagnosis. Not a certain one.

Every affected person had some developmental delay.
Common features included delayed speech and autism.



Families who can't pay can ask.

Candidates are underserved people with a suspected rare disease who can't pay for sequencing. Testing and follow-up are free, 5 visits in 2 years, in person or virtual. Write to texome-project@bcm.edu.

